Forensic Analysis and Recovery of Corrupted MBR Disk

Prepared by: Shahad Alharbi

Tools Used: 010 Editor, FTK Imager

1. Incident Description

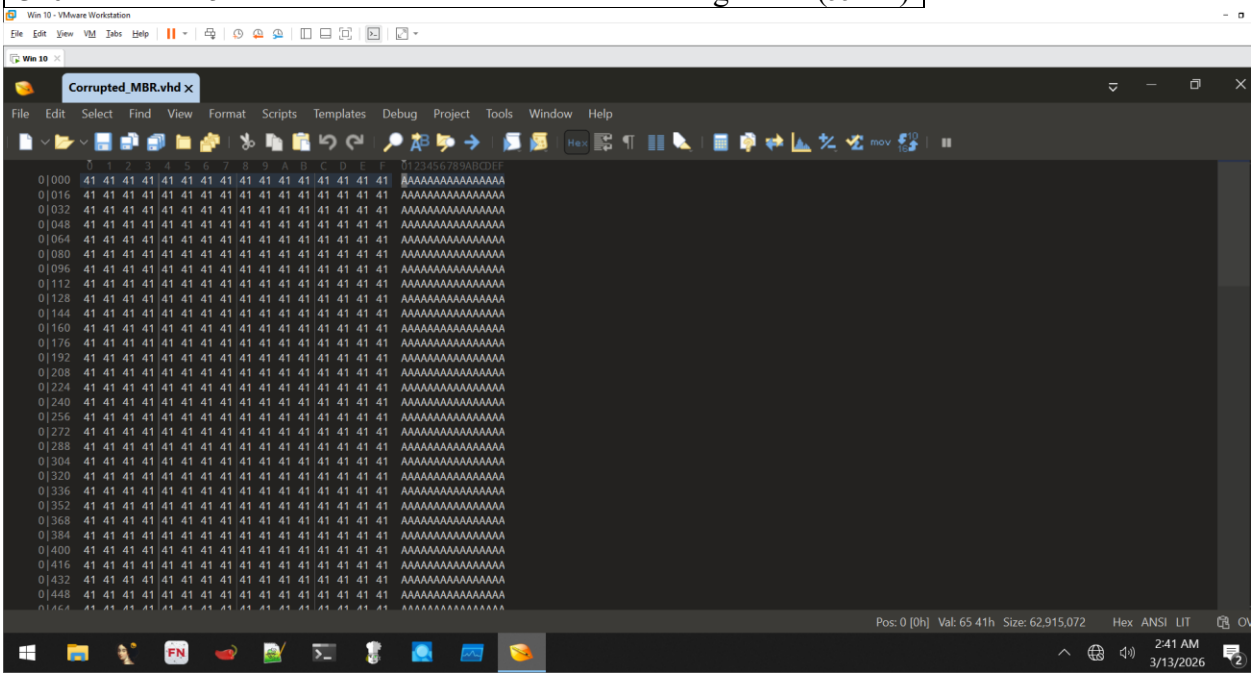
In this scenario, two NTFS partitions were initially created and populated with sample evidence files. One partition contained an image file intended to represent suspicious visual content a cat image used as simulated evidence), while the second partition contained a text file with a message suggesting deliberate concealment of information.

To simulate anti-forensic activity, the first 1024 bytes of the disk were intentionally overwritten. This corruption targeted the beginning of the disk to prevent normal partition recognition and make the stored evidence inaccessible through conventional forensic tools.
This corruption was destroyed:

- Master Boot Record (MBR)
- Partition table
- Beginning of first partition boot area

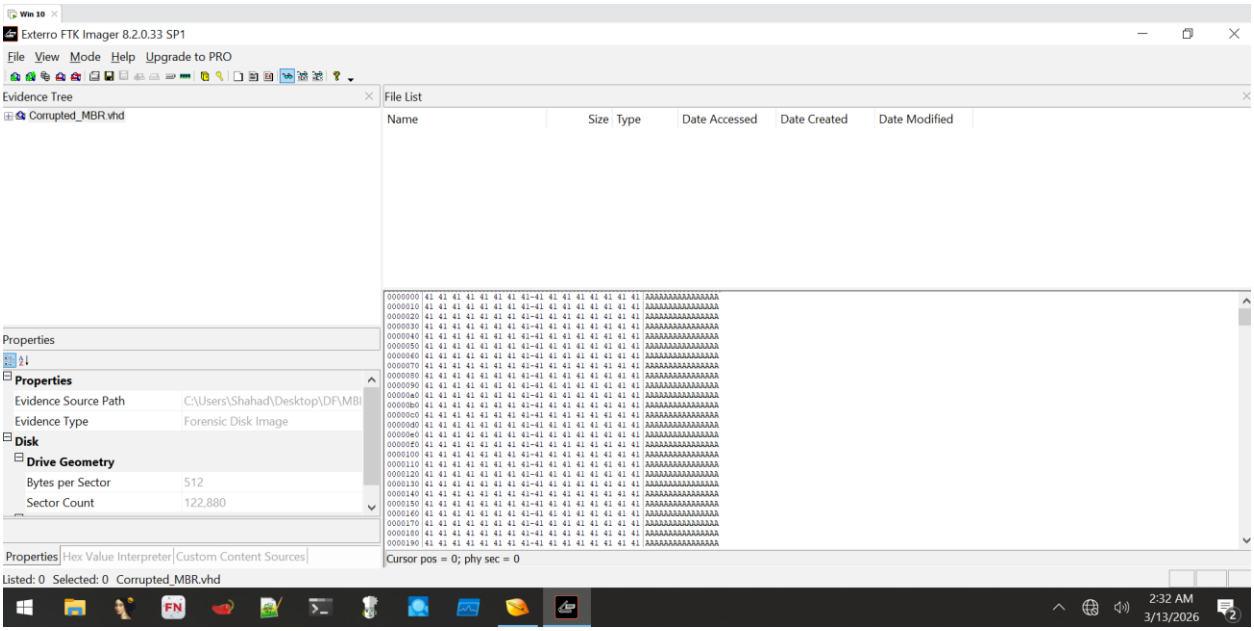
Original MBR Structure (Sector 0)

Offset Start	Offset End	Length (Bytes)	Description
0	445	446	Boot Code Area
446	509	64	Master Partition Table
510	511	2	Boot Signature (55 AA)



2. Initial Disk Examination

The corrupted disk image was opened using FTK Imager.



The initial result showed:

Unrecognized file system [unknown type]

This confirmed that:

- The original MBR was no longer valid
- Partition entries were destroyed
- File system references could not be read automatically.

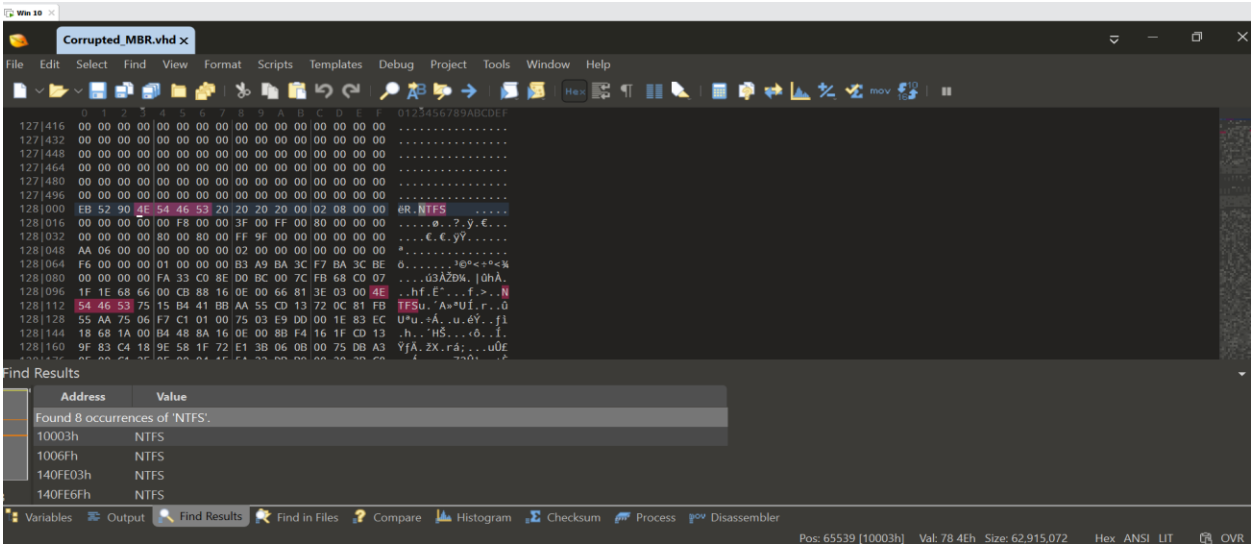
3. Recovery Strategy

Because the partition table was missing, the recovery process relied on locating surviving NTFS structures manually in case of a disk using NTFS filesystem.

The keyword searched was **NTFS**

Hex signature: **4E 54 46 53**

This signature is located inside every NTFS Volume Boot Record (VBR).



The search returned 8 occurrences. This became the key indicator for locating the surviving partitions.

Each NTFS partition normally contains:

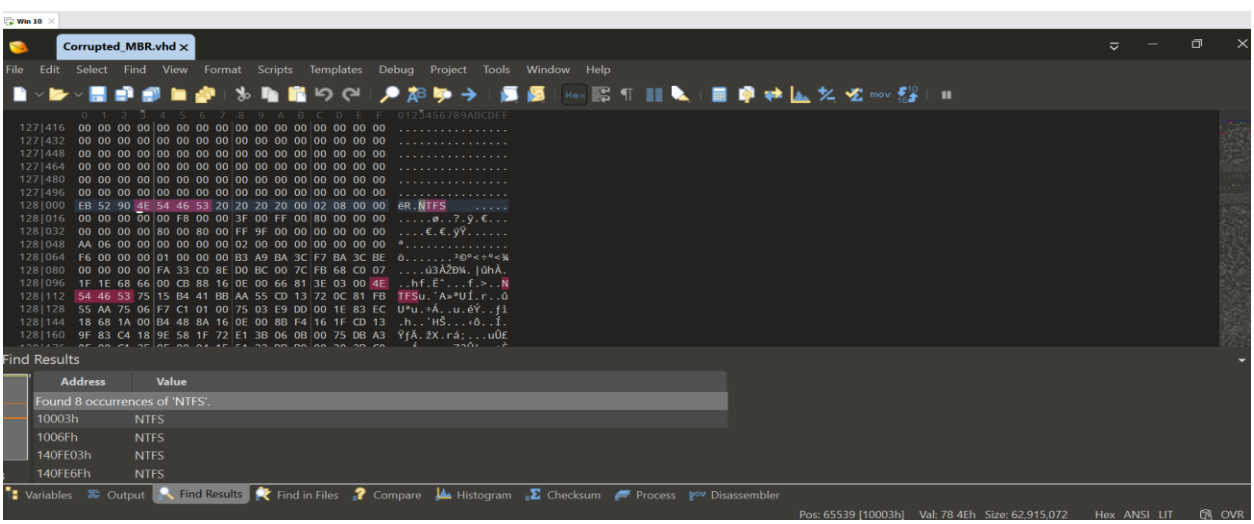
- one primary Volume Boot Record
- one backup boot sector

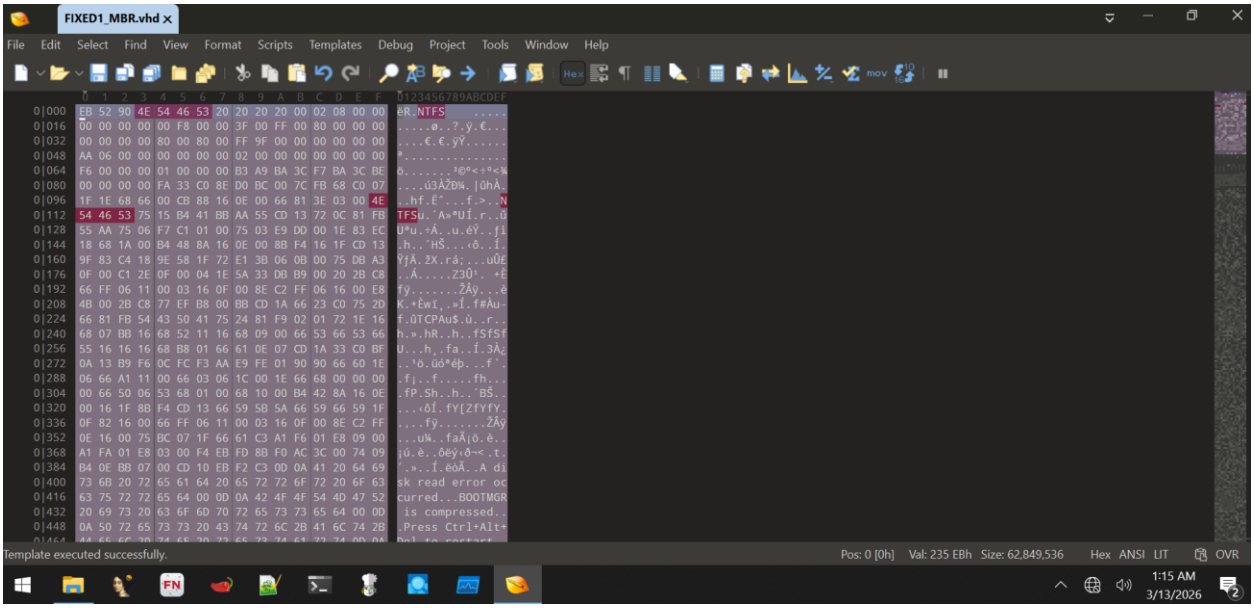
Within each of these sectors, the NTFS signature appears twice in nearby positions. Therefore, eight NTFS signatures were found.

4. Manual Reconstruction of the First Partition

Because the MBR was no longer usable, the corrupted beginning of the disk was logically ignored.

The disk was treated directly as a volume beginning at: 10000h



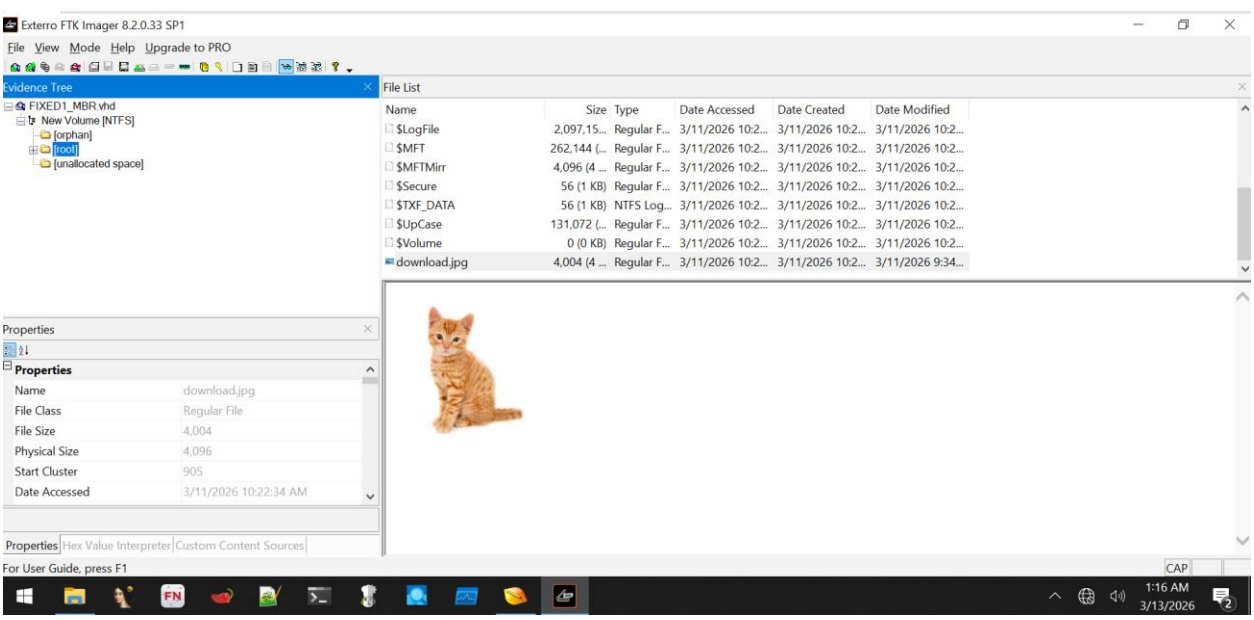


5. Recovery and Validation of First Partition Evidence

After reconstruction, FTK Imager successfully displayed the first partition contents.

Recovered files included: download.jpg

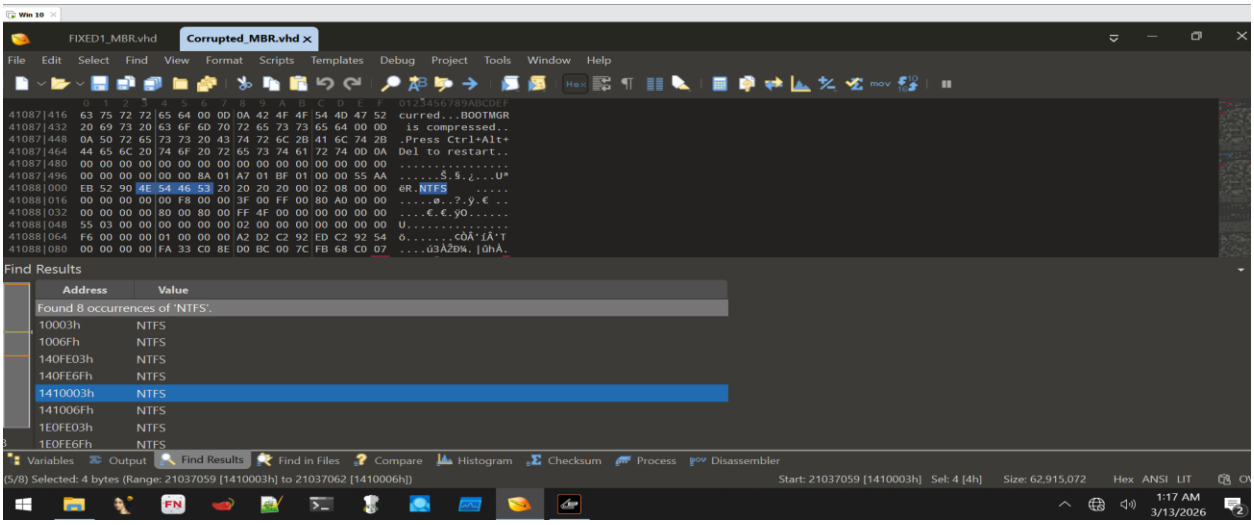
The image file opened successfully, confirming that the partition data remained readable despite the destruction of the original MBR.



The recovered image showed suspicious content, confirming successful file level recovery.

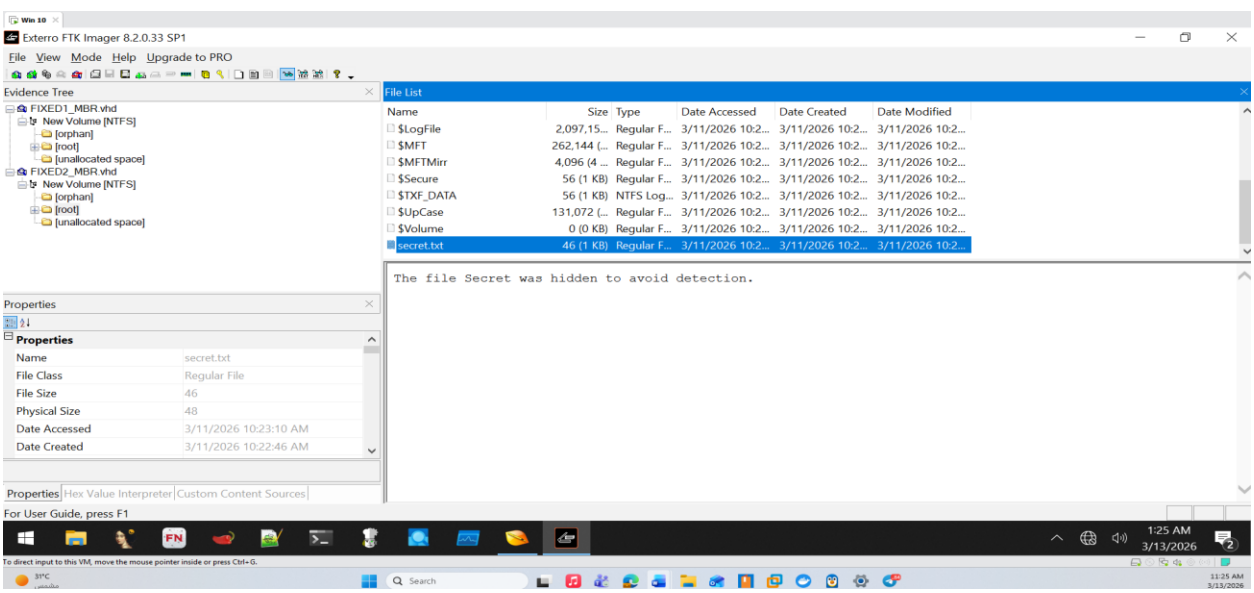
6. Manual Reconstruction of the Second Partition

The same reconstruction method used for the first partition was repeated to recover the second NTFS volume. During NTFS signature analysis, the first four NTFS occurrences were associated with the first partition because they represented the primary and backup boot sector structures of that volume. To locate the second partition, attention was shifted to the fifth NTFS occurrence, which indicated the beginning of the next surviving NTFS structure within the disk image.



7. Evidence Recovered from the Second Partition

Recovered files from the second partition included: secret.txt



This confirms that meaningful user-created evidence remained recoverable after disk corruption.

8. Why Recovery Was Possible Despite MBR Destruction

Recovery remained possible because the MBR only stores partition references.

The actual NTFS file systems begin deeper inside the disk and contain their own boot sectors and metadata.

Therefore:

Even if the MBR is destroyed:

- VBR survives
- NTFS metadata survives
- files remain recoverable